

VEDAANG CHOPRA

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MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

AI Scientist: Production ML (4.5 yrs, enterprise scale) – Causal/ablation evaluation – LLM/GenAI applied research

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Built and deployed production ML for enterprise network telemetry: SLA forecasting pipelines serving **60+ classifiers** across 4 categories with automated model evaluation and retraining; unsupervised anomaly detection and data mining over large-scale event data (patent PCT/IN2022/058026; reduced manual troubleshooting **~75%**).
- Owned end-to-end ML pipelines from data acquisition to model output delivery: scaled ingestion from 50 to **2,000 events/sec (40x)** in Golang — discovering sources, building ETL and featurization, and making large-scale telemetry model-ready with reliability and observability.
- Led an agentic RAG diagnostics system from hackathon prototype to production: LLM plans multi-step tool calls over telemetry and verifies hypotheses via structured function calling — reduced mean resolution time **~70%**.
- Deployed inference-optimized models via ONNX Runtime on edge appliances (**~40%** latency reduction); integrated ML into Python/Go backend services across distributed infrastructure.

RESEARCH EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Designing execution-grounded evaluation for LLM code generation: geometric verification (Chamfer/Hausdorff on STL/B-Rep), compile validity, and VLM-based visual judgment — maintained as a **202-test harness** across a 16-module system framing synthesis as inference-time search with verifiable reward.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **ATHENA**: Built a multi-agent generation system (**5 specialized agents**, supervisor-worker orchestration via LangGraph Command) with reflection-based quality critique — ablation studies showed removing the critique agent cut fact coverage by **27%**; evaluated against **100 reference videos** with BLEU-4 (+0.18 absolute) and CLIPScore.
- **ARTEMIS**: Developing cost-aware routing that selects the optimal Vision-Language Model per request — trained neural router, SLA-aware load balancing, **6+ backends served via vLLM** across VQA, OCR, captioning, and multimodal reasoning tasks.
- **CERBERUS**: Cross-modal alignment of frozen encoders (CLIP/SBERT) with Matryoshka embedding compression (4096→128 dims, **~96%** retention); trained with PyTorch DDP + mixed precision on H100/A100 clusters; achieved **R@5 78%** on PixMo retrieval.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (Machine Learning), GPA: 4.0/4.0

Aug 2024 – Dec 2026

- **Graduate Coursework**: Deep Learning, Deep Reinforcement Learning, ML Security, Large & Vision Language Models, Agentic AI, Systems for AI — covering supervised/unsupervised learning, reinforcement learning, Bayesian approaches, and experimental design & analysis.

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless anomaly detection.

IEEE ICAIA 2026 — ML + Memory Forensics for Ransomware Detection in IoT-Enabled Energy Systems.

TECHNICAL SKILLS

ML Paradigms: Supervised/Unsupervised Learning, Deep Learning, A/B Test Design & Analysis, Ablation Studies, Data Mining, Model Evaluation & Benchmarking, Experimentation & Success-Criteria Definition, Explainability (ablation- and attribution-based)

Deep Learning / NLP: PyTorch, Hugging Face Transformers, Scikit-Learn, NLP (BLEU, METEOR, BERTScore, CLIPScore), Computer Vision, Vision-Language Models

LLMs / Generative AI: LangChain, LangGraph, Agentic RAG, Tool/Function Calling, Structured Generation, Prompt Evaluation, vLLM, Inference Optimization, LLM-as-a-Judge

Data & Systems: Python, Go, SQL/PostgreSQL, OpenSearch/Elasticsearch, Pandas, NumPy, Docker, Linux, Git, FastAPI, Azure

Reinforcement Learning: PPO, DQN, Reward Shaping, Curriculum Learning, Self-Play (academic projects)